

AUTO MERGE AND MANUAL MERGE

A: AUTOMATIC MERGE - FAST FORWARD

```
S DEEKSHA@LAPTOP-1FDIVFOE MINGW64 ~  
$ cd /d/  
mkdir git-auto-merge-demo  
cd git-auto-merge-demo  
git init  
git config user.name "S75Deeksha"  
git config user.email "SriDeeksha75Acharya@gmail.com"  
Initialized empty Git repository in D:/git-auto-merge-demo/.git/  
  
S DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-auto-merge-demo (master)  
$
```

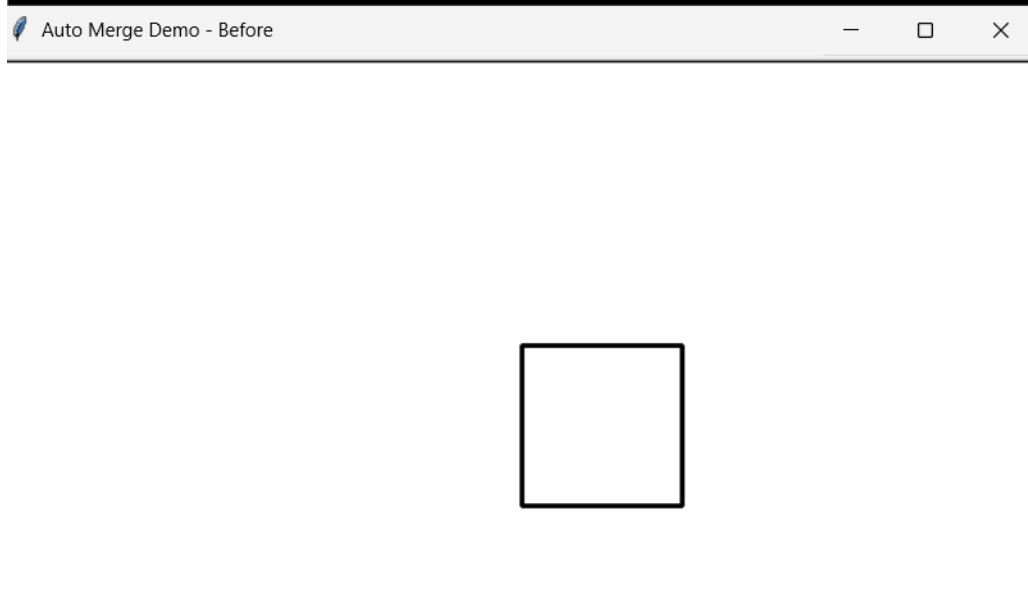
Initializing git-auto-merge-demo repository

```
$ DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-auto-merge-demo (master)
$ vi draw_square.py

$ DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-auto-merge-demo (master)
$ git add draw_square.py
warning: in the working copy of 'draw_square.py', LF will be replaced by CRLF
the next time Git touches it

$ DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-auto-merge-demo (master)
$ git commit -m "Initial: Draw basic square"
[master (root-commit) e6cb044] Initial: Draw basic square
1 file changed, 20 insertions(+)
create mode 100644 draw_square.py

$ DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-auto-merge-demo (master)
$ py draw_square.py
```



```
$ DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-auto-merge-demo (master)
$ git log --oneline
e6cb044 (HEAD -> master) Initial: Draw basic square
```

Initial commit on main + plain square output

```
MINGW64:/d/git-auto-merge-demo
Auto Merge Demo - After Merge

Auto Merge Success!

$ DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-auto-merge-demo (master)
$ git checkout -b feature/add-color
Switched to a new branch 'feature/add-color'

$ DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-auto-merge-demo (feature/add-color)
$ vi draw_square.py

$ DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-auto-merge-demo (feature/add-color)
$ git add draw_square.py
warning: in the working copy of 'draw_square.py', LF will be replaced by CRLF
the next time Git touches it

$ DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-auto-merge-demo (feature/add-color)
$ git commit -m "Feature: Add red border, yellow fill, blue background"
[feature/add-color 89daedc] Feature: Add red border, yellow fill, blue backgr
ound
1 file changed, 13 insertions(+), 4 deletions(-)

$ DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-auto-merge-demo (feature/add-color)
$ git log --oneline --all --graph
* 89daedc (HEAD -> feature/add-color) Feature: Add red border, yellow fill, b
lue background
* e6cb044 (master) Initial: Draw basic square

$ DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-auto-merge-demo (feature/add-color)
$ py draw_square.py
File "D:\git-auto-merge-demo\draw_square.py", line 29
    draw_colored_square()
                        ^
SyntaxError: unmatched ')'
```

```

$ DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-auto-merge-demo (feature/add-color)
$ git add draw_square.py
warning: in the working copy of 'draw_square.py', LF will be replaced by CRLF
the next time Git touches it

$ DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-auto-merge-demo (feature/add-color)
$ git commit -m "did some changes in the program"
[feature/add-color 43d568a] did some changes in the program
1 file changed, 3 insertions(+), 3 deletions(-)

$ DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-auto-merge-demo (feature/add-color)
$ git log --oneline --all --graph
* 43d568a (HEAD -> feature/add-color) did some changes in the program
* 89daedc Feature: Add red border, yellow fill, blue background
* e6cb044 (master) Initial: Draw basic square

```

Some commits

```

$ DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-auto-merge-demo (feature/add-color)
$ git checkout main
error: pathspec 'main' did not match any file(s) known to git

$ DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-auto-merge-demo (feature/add-color)
$ git branch
* feature/add-color
  master

$ DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-auto-merge-demo (feature/add-color)
$ git checkout master
Switched to branch 'master'

$ DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-auto-merge-demo (master)
$ git merge feature/add-color
Updating e6cb044..43d568a
Fast-forward
 draw_square.py | 17 ++++++++-----
1 file changed, 13 insertions(+), 4 deletions(-)

$ DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-auto-merge-demo (master)
$

```

Terminal showing Fast-forward merge

```
S DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-auto-merge-demo (master)
$ git add README.md
git commit -m "Add README for automatic merge"
warning: in the working copy of 'README.md', LF will be replaced by CRLF the
next time Git touches it
[master a5b8315] Add README for automatic merge
1 file changed, 14 insertions(+)
create mode 100644 README.md

S DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-auto-merge-demo (master)
$ git remote add origin "https://github.com/S75Deeksha/auto_merge.git"

S DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-auto-merge-demo (master)
$ git push -u origin master
Enumerating objects: 12, done.
Counting objects: 100% (12/12), done.
Delta compression using up to 12 threads
Compressing objects: 100% (9/9), done.
Writing objects: 100% (12/12), 1.65 KiB | 562.00 KiB/s, done.
Total 12 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/S75Deeksha/auto_merge.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.

S DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-auto-merge-demo (master)
$
```

Push is done

B: Manual Merge - Conflict Resolution

```
S DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d
$ mkdir git-manual-merge-demo
  cd git-manual-merge-demo
  git init
  git config user.name "S75Deeksha"
  git config user.email "SriDeeksha75Acharya@gmail.com"
Initialized empty Git repository in D:/git-manual-merge-demo/.git/
```

Initializing git-manual-merge-demo repository

```
$ DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-manual-merge-demo (master)
$ vi draw_star.py

$ DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-manual-merge-demo (master)
$ git add draw_star.py
git commit -m "Initial: Draw white star size 100"
py draw_star.py
warning: in the working copy of 'draw_star.py', LF will be replaced by CRLF t
he next time Git touches it
[master (root-commit) 61c736b] Initial: Draw white star size 100
1 file changed, 21 insertions(+)
create mode 100644 draw_star.py
```

Manual Merge Demo - Star Size 100



White star size 100 output

```
Manual Merge Demo - Star Size 100

$ DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-manual-merge-demo (master)
$ git checkout -b feature/big-green-star
vi draw_star.py
Switched to a new branch 'feature/big-green-star'

$ DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-manual-merge-demo (feature/big-green-star)
$ git add draw_star.py
git commit -m "Feature: Change star to size 200 and lime color"
py draw_star.py
git log --oneline
warning: in the working copy of 'draw_star.py', LF will be replaced by CRLF the next time Git touches it
[feature/big-green-star a26ab38] Feature: Change star to size 200 and lime color
1 file changed, 1 insertion(+), 1 deletion(-)
a26ab38 (HEAD -> feature/big-green-star) Feature: Change star to size 200 and lime color
61c736b (master) Initial: Draw white star size 100
```

Big lime star size 200 output

```

$ DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-manual-merge-demo (feature/big-green
-star)
$ git checkout master
vi draw_star.py
Switched to branch 'master'

$ DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-manual-merge-demo (master)
$ git add draw_star.py
git commit -m "Main: Change star to size 50 and cyan color"
git log --oneline --graph --all
[master 9ec5ac4] Main: Change star to size 50 and cyan color
1 file changed, 1 insertion(+), 1 deletion(-)
* 9ec5ac4 (HEAD -> master) Main: Change star to size 50 and cyan color
| * a26ab38 (feature/big-green-star) Feature: Change star to size 200 and lim
e color
|/
* 61c736b Initial: Draw white star size 100

$ DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-manual-merge-demo (master)
$

```

Diverged graph - both branches changed same line

```

$ DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-manual-merge-demo (master)
$ git merge feature/big-green-star
Auto-merging draw_star.py
CONFLICT (content): Merge conflict in draw_star.py
Automatic merge failed; fix conflicts and then commit the result.

$ DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-manual-merge-demo (master|MERGING)
$

```

Terminal CONFLICT merge conflict in draw_star.py


```

$ DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-manual-merge-demo (master|MERGING)
$ cat draw_star.py
import turtle

def draw_star(size, color):
    screen = turtle.Screen()
    screen.bgcolor("black")
    screen.title(f"Manual Merge Demo - Star Size {size}")

    t = turtle.Turtle()
    t.speed(3)
    t.color(color)
    t.pensize(2)

    for _ in range(5):
        t.forward(size)
        t.right(144)

    t.hideturtle()
    screen.exitonclick()

if __name__ == "__main__":
<<<<<<< HEAD
    draw_star(50, "cyan")
=====
    draw_star(100, "lime")
>>>>>>> feature/big-green-star

$ DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-manual-merge-demo (master|MERGING)
$ 

```

File with <<<<<<< HEAD size=50,cyan =====
size=200,lime >>>>>>>

```

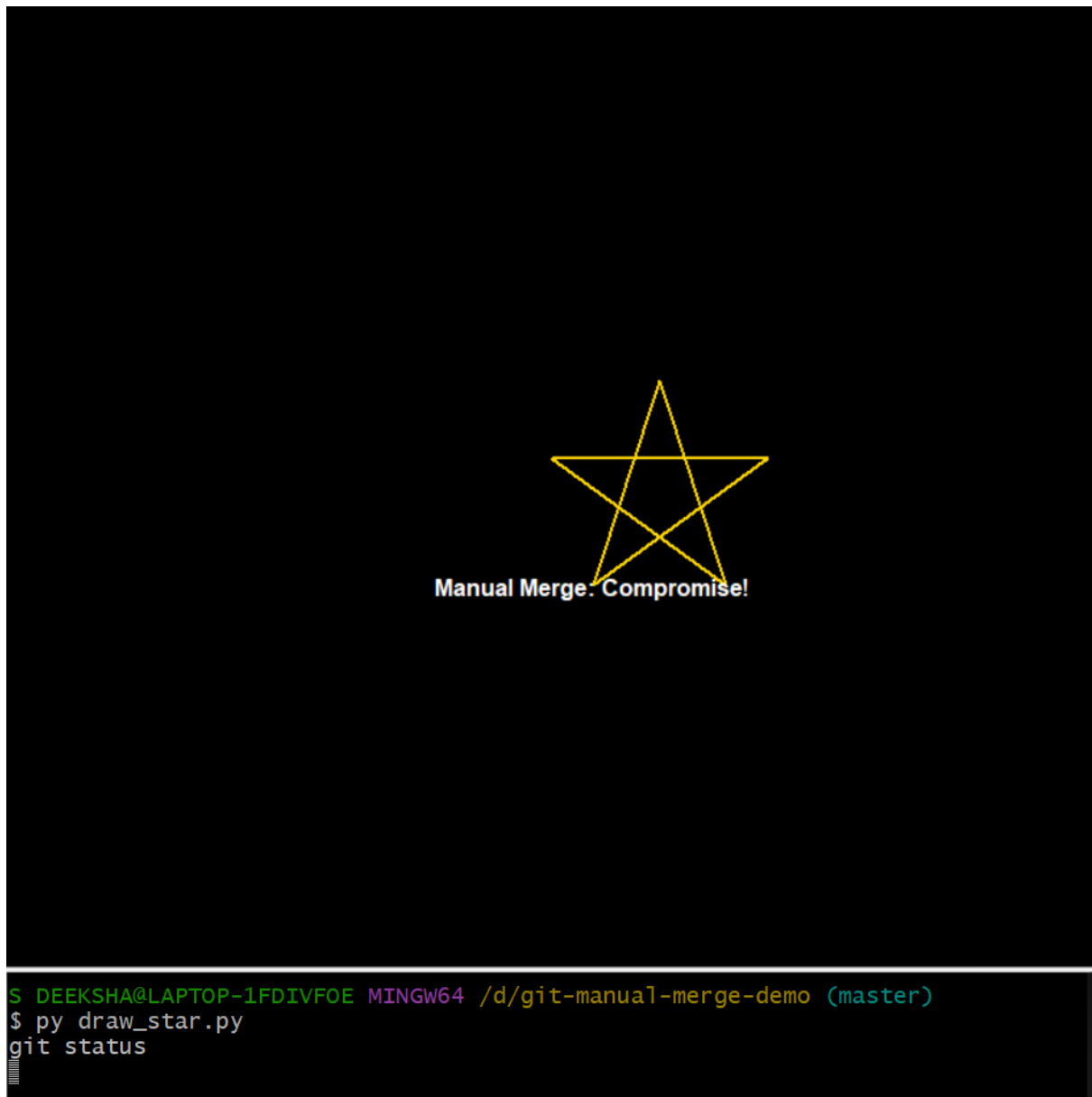
$ DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-manual-merge-demo (master|MERGING)
$ vi draw_star.py

$ DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-manual-merge-demo (master|MERGING)
$ git add draw_star.py
git commit -m "Resolve: Manual merge - compromise size 150, gold color"
git log --oneline --graph
[master 8d0f8ca] Resolve: Manual merge - compromise size 150, gold color
*   8d0f8ca (HEAD -> master) Resolve: Manual merge - compromise size 150, gold color
|\
| * a26ab38 (feature/big-green-star) Feature: Change star to size 200 and lime color
* | 9ec5ac4 Main: Change star to size 50 and cyan color
|/
* 61c736b Initial: Draw white star size 100

$ DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-manual-merge-demo (master)
$ 

```

Merge commit graph with 2 parents



Gold star size 150 + "Manual Merge: Compromise!"
text

```

S DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-manual-merge-demo (master)
$ vi README.md

S DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-manual-merge-demo (master)
$ git add README.md
git commit -m "Add README for manual merge"
warning: in the working copy of 'README.md', LF will be replaced by CRLF the
next time Git touches it
[master 72b87ff] Add README for manual merge
1 file changed, 15 insertions(+)
create mode 100644 README.md

S DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-manual-merge-demo (master)
$ cat README.md
# 📄 Manual Merge Demo - Conflict Resolution

## Why Manual Merge?
Both branches changed `draw_star(50, "cyan")` vs `draw_star(200, "lime")` on
SAME LINE. Git: "You decide."

## The Battle
## Human Victory
1. Git added `<<<<<< HEAD` markers
2. We chose `draw_star(150, "gold")` - compromise
3. `git add` + `git commit` sealed merge

## Run the Peace Treaty
...bash
python draw_star.py
e

S DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-manual-merge-demo (master)
$

```

README.md content

```

S DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-manual-merge-demo (master)
$ git remote add origin "https://github.com/S75Deeksha/manual_merge.git"

S DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-manual-merge-demo (master)
$ git push -u origin master
Enumerating objects: 15, done.
Counting objects: 100% (15/15), done.
Delta compression using up to 12 threads
Compressing objects: 100% (11/11), done.
Writing objects: 100% (15/15), 1.82 KiB | 310.00 KiB/s, done.
Total 15 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (4/4), done.
To https://github.com/S75Deeksha/manual_merge.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.

S DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-manual-merge-demo (master)
$

```

```
S DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-manual-merge-demo (master)
$ git remote add origin "https://github.com/S75Deeksha/manual_merge.git"

S DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-manual-merge-demo (master)
$ git push -u origin master
Enumerating objects: 15, done.
Counting objects: 100% (15/15), done.
Delta compression using up to 12 threads
Compressing objects: 100% (11/11), done.
Writing objects: 100% (15/15), 1.82 KiB | 310.00 KiB/s, done.
Total 15 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (4/4), done.
To https://github.com/S75Deeksha/manual_merge.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.

S DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-manual-merge-demo (master)
$
```

Pushed to git

USING MERGING TOOL

[illegible]

```
$ DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-manual-merge-demo (mergetool-demo|ME
$ git status
On branch mergetool-demo
All conflicts fixed but you are still merging.
  (use "git commit" to conclude merge)

Changes to be committed:
  modified:   draw_star.py

Untracked files:
  (use "git add <file>..." to include in what will be committed)
  draw_star.py.orig

$ DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-manual-merge-demo (mergetool-demo|ME
$ git commit -m "Resolve: Used vimdiff mergetool - compromise size 150, gold"
[mergetool-demo bb5cf46] Resolve: Used vimdiff mergetool - compromise size 15

$ DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-manual-merge-demo (mergetool-demo)
$ py draw_star.py
```

MINGW64:/d/git-manual-merge-demo

```
$ git mergetool
```

```
Merging:  
draw_star.py
```

```
Normal merge conflict for 'draw_star.py':  
  {local}: modified file  
  {remote}: modified file  
4 files to edit
```

```
$ DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-manual-merge-demo (master)
$ git config --global merge.tool vimdiff
git config --global mergetool.prompt false

$ DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-manual-merge-demo (master)
$ git merge --abort
git merge feature/big-green-star
fatal: There is no merge to abort (MERGE_HEAD missing).
Already up to date.

$ DEEKSHA@LAPTOP-1FDIVFOE MINGW64 /d/git-manual-merge-demo (master)
$ git mergetool
```

